Active-layer thickness prediction and regional differences across permafrost regions using a physics-residual machine-learning workflow

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Abstract

Active-layer thickness (ALT) varies from year to year, yet gridded products are mostly trained on multi-year conditions and rarely evaluated against same-location, same-year observations. We combined 19,172 site-year observations from Alaska, the Lena Delta and north-western Canada (2010–2024) with ERA5-Land thawing degree-days accumulated to the observation day, and tested a physics-residual workflow in which gradient boosting learns the residual of a locally recalibrated Stefan model. Under spatial-block cross-validation, the physics-residual model reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada and matched it in Alaska; the gain required soil and terrain covariates and was largest where the regional Stefan model erred most. When the Lena Delta was withheld from training, direct machine learning exceeded the Stefan error by 17.8 cm, whereas the physics-residual model remained close to it, and 25 local labels then reduced its error from 23.8 to 19.9 cm. In the same cells and years, the Lena Delta error (21.1 cm) was lower than that of five published products (28.4–34.6 cm), which do not represent its thin active layers. The workflow provides year-labelled 1 km ALT maps of the three regions for 2011–2024 with an area of applicability and prediction intervals.

Keywords: permafrost, active-layer thickness, Stefan model, physics-guided machine learning, spatial cross-validation, ERA5-Land

Introduction

The active layer is the near-surface ground above permafrost that thaws each summer and refreezes each winter. Its thickness (ALT) sets the depth of seasonal thaw and controls ground stability, near-surface hydrology and the amount of soil organic carbon exposed to decomposition[1–3]. ALT responds to the thermal regime of individual years, so that interannual variability is an intrinsic part of the signal recorded by long-term monitoring programmes such as the Circumpolar Active Layer Monitoring (CALM) network[4,5]. Observations, however, are sparse and clustered: most measurements originate from a small number of intensive field campaigns and long-term grids, and many sites are visited only once[6,7].

Several gridded ALT products at about 1 km resolution now cover the northern permafrost region. Process-based estimates such as ESA CCI Permafrost solve the ground heat equation forced by remotely sensed land-surface temperature[8,9], whereas statistical and machine-learning (ML) products relate site observations to climate, terrain and soil predictors[10–13]. Three limitations restrict their use for year-specific applications. First, most products are trained on multi-year mean predictors or on site-years pooled across many years, so that a product value is not tied to the climate of a stated year. Second, products are evaluated with different site sets and validation schemes; reported errors range from a random ten-fold root-mean-square error (RMSE) of 21.6 cm[12] to a leave-one-site-out RMSE of 60.5 cm[13] and cannot be compared with each other. Third, random cross-validation of spatially clustered observations overstates the accuracy of ML models at new locations[14–17], which hides how a product performs in a region without observations, that is, where it is most needed.

Physically based models offer a complementary route. The Stefan solution relates thaw depth to the thawing degree-days (TDD) of the season through a single coefficient that depends on soil thermal properties and moisture[4,18,19]. It reproduces the seasonal progression of thaw and responds to the climate of each year, but one coefficient per region leaves systematic residuals associated with soil organic matter and terrain. Physics-guided ML, in which a physical model carries the dominant response and a learner corrects what the physics omits, has improved accuracy and extrapolation in several environmental applications[20–22], and in Alaska the Stefan model retained its accuracy on held-out sites better than a random forest[23]. Whether such a combination improves on the Stefan model when the climate input is matched to the observation year, whether it transfers to a region without labels, how many local labels a new region requires, and how it compares with the published products in the same cells and years has not been tested.

Here we address these questions with a site-year design, in which each ALT observation is an observation site in a stated year and each climate input is computed from ERA5-Land for that year and accumulated to the observation day[24]. Using 19,172 site-year observations from 2010–2024 in Alaska, the Lena Delta and north-western Canada, we compare the Stefan model, direct ML and a physics-residual workflow in which gradient boosting learns the residual of a recalibrated Stefan model. All methods and five published products are scored against the same observations under the same validation designs, including the hold-out of an entire region and a test in eight external CALM regions, which is what makes their accuracies comparable. The study

(i) quantifies the effect of the year-specific climate input, (ii) identifies the conditions under which a learner improves on the physics and the conditions under which the physical anchor preserves transferability, (iii) establishes how many local labels a new region needs, and (iv) provides year-labelled 1 km ALT maps for 2011–2024 with an area of applicability and prediction intervals, together with the evaluation protocol.

Data and methods

Study regions and active-layer observations

We used three regions with dense, multi-year thaw-depth observations: Alaska, the Lena Delta and north-western Canada (Canada (NW) in the figures and tables; Fig. 1, Table 1). Observations for Alaska and Canada were taken from the ABoVE soil moisture and active-layer thickness compilation, version 2[6], and from the CALM end-of-season grid values[5]; observations for the Lena Delta were taken from the ALLena thaw-depth data set[7]. ABoVE probe measurements were restricted to August–October and to $0 < \mathrm{ALT} < 300$ cm, because measurements made earlier in the season record thaw depth rather than the seasonal maximum. ALLena measurements were restricted to August–September values without the probe-limit flag; the seasonal maximum of each location and year was taken as the label and the mean day of its measurements as the observation day. Observations were grouped by site, defined as a location with coordinates rounded to 0.0001° (about 10 m), and averaged within each site and year, retaining the mean day of observation; probe points of the ABoVE transects are therefore separate sites, and several sites can lie in one 1 km cell. CALM values without a date were assigned day 244 (1 September). We kept the years 2010–2024, for which year-specific climate inputs are available. The resulting labels comprise 15,540 site-years in Alaska (2010–2024), 2,436 in the Lena Delta (2010–2022) and 1,196 in Canada (2010–2019 and 2021–2024), distributed over 73, 16 and 26 blocks of 0.5° (Table 1). The sites lie in 321, 143 and 71 cells of 1 km, that is 910, 244 and 144 cell-years; up to 701 sites of one transect fall in one cell and year, so that site-years are not independent samples at the 1 km scale, which the spatial-block design accounts for. Sources, versions and licences are listed in Supplementary Table S1, the years used in each analysis in Supplementary Table S2 and the processing steps in Supplementary Methods S1.

Year-specific climate input

Monthly ERA5-Land fields for January 2010 to December 2024[24] (2 m air temperature, top-layer soil temperature and snow water equivalent) were extracted at the nearest land grid cell of each site. For site c and year y, the thawing degree-day sum is

$$\mathrm{TDD}_{c,y} = \sum_{m=1}^{12} \max(T_{c,y,m},\, 0)\, d_{y,m}, \tag{1}$$

where $T_{c,y,m}$ is the monthly mean air temperature and $d_{y,m}$ the number of days in month m. Because most observations were made before the end of the thaw season,

Table 1 Data and years used. Top: ALT labels by region (site-years of 2010–2024 with year-specific climate input; a site is an observation location with coordinates rounded to 0.0001°, and the 1 km cells are the map cells that contain the sites). Bottom: inputs and products, with the period available and the years used in this study. Blocks are the 0.5° cells used for spatial cross-validation and bootstrap; there are no Canadian labels in 2020.

Region	Site-years	Sites	1 km cells	Blocks	Years	Sites ≥4 yr	Largest year
Alaska	15,540	13,253	321	73	2010–2024	203	2014 (51%)
Lena Delta	2,436	2,206	143	16	2010–2022	25	2016 (28%)
Canada (NW)	1,196	730	71	26	2010–2024	48	2022 (37%)
External CALM	805	72	–	49	2010–2024	–	–

Data	Resolution	Available	Used	Role
ERA5-Land monthly [24]	0.1°	1950–	2010–2024; mean 2011–2024	Climate input
Copernicus DEM [25]	30 m	static	–	Terrain
SoilGrids 2.0 [26]	250 m	static	–	Soil
MODIS MOD13Q1, MOD10A2 [27,28]	250–500 m	2000–	2010–2024	Sensitivity test
ESA CCI permafrost probability [8]	1 km	1997–2021	mean	Map domain
Wei et al. [13]	1 km	2000–2024	2010–2024	Comparison
Liu et al. [12]	1 km	2003–2020	2010–2020	Comparison
ESA CCI ALT v4 [8]	1 km	1997–2021	2010–2021	Comparison
ESA CCI ALT v5 [9]	1 km	1997–2023	2010–2023	Comparison
Yi and Kimball [29]	1 km	2001–2015	2010–2015 (Alaska)	Reference, not scored
Aalto et al. [10]	30″	2000–2014	time-invariant	Comparison

the model input is the TDD accumulated to the day of observation t,

$$\mathrm{TDD}^{\mathrm{obs}}_{c,y} = C_{c,y,k-1} + \frac{t - t_{k-1}}{d_{y,k}} \left(C_{c,y,k} - C_{c,y,k-1} \right), \tag{2}$$

where $C_{c,y,k}$ is the cumulative TDD to the end of month k, k is the month containing t and t_{k-1} is the last day of the preceding month. Seven further climate variables were computed for each cell and year: the mean annual air temperature, $\sqrt{\mathrm{TDD}}$, the warmest and the coldest monthly mean temperature, the freezing degree-days of the preceding winter (October $y-1$ to April y), the mean top-layer soil temperature and the mean snow water equivalent of the hydrological year (October $y-1$ to September y) (Supplementary Table S3, Supplementary Methods S2). Two alternative inputs were evaluated on the same site-years: the full-season TDD of the observation year, and the 2011–2024 mean of the yearly values (mean climate input), which corresponds to the multi-year inputs of existing products. Yearly maps use the TDD accumulated to day of year 244 (1 September in non-leap years), the end-of-season reference date of the CALM protocol; this input lies within the range of the training labels (pooled interquartile range of observation days 224–234), whereas the full-season sum exceeds

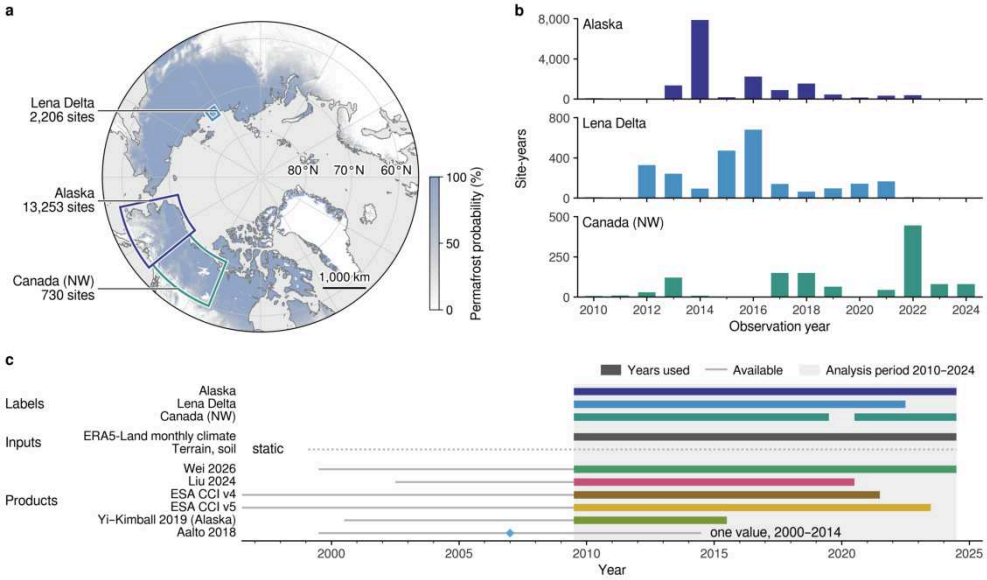


Fig. 1 Study regions, observations and data periods. a, ESA CCI permafrost probability (mean of 1997–2021) with the three study regions and the number of sites with ALT labels in 2010–2024. **b**, Site-year labels per observation year. **c**, Periods covered by the labels, the ERA5-Land climate input, the static covariates and the six published products; dark bars mark the years with product values at the label sites, and the shaded band marks the analysis period.

the observation-day sum by 12–37% depending on the region. Yearly map values therefore represent ALT on 1 September.

Static covariates

Six terrain variables (elevation, slope, the sine and cosine of aspect, topographic position index and roughness) were computed from the 30 m Copernicus DEM[25] in a 33×33 pixel window (about 1 km) around each site. Nine soil variables (clay, sand, silt, bulk density, coarse fragments and pH at 5–15 cm, and soil organic carbon at 0–5, 5–15 and 15–30 cm) were sampled from SoilGrids 2.0 at 250 m[26]. The main feature vector $\mathbf{x}$ therefore contains 23 variables (eight climate, six terrain and nine soil). Sensitivity tests used nested sets: climate only; climate and terrain; the main set plus MODIS summer NDVI, EVI and snow-off timing for 2010–2024 (MOD13Q1 and MOD10A2, version 6.1)[27,28]; the main set plus high-resolution soil carbon and land-cover classes[30]; and all variables including the climate of the preceding year. The ESA CCI permafrost probability (mean of 1997–2021)[8] was used only to define the map domain (probability of at least 50%) and not as a predictor.

Model formulation

The Stefan model describes the thaw depth as

$$\widehat{\mathrm{ALT}}_{\mathrm{S}} = E\sqrt{\mathrm{TDD}^{\mathrm{obs}}}, \tag{3}$$

where E ($\mathrm{cm}\,(^{\circ}\mathrm{C}\,\mathrm{d})^{-1/2}$) summarises the thermal properties and moisture of the soil[18,19]. The relation holds within a season at the Lena Delta monitoring sites and, with more scatter, among years at cells observed repeatedly (Supplementary Fig. S1). For a set of n training labels y_i with $s_i = \sqrt{\mathrm{TDD}_i^{\mathrm{obs}}}$, the coefficient is recalibrated by least squares through the origin,

$$\hat{E} = \frac{\sum_{i=1}^{n} y_i\, s_i}{\sum_{i=1}^{n} s_i^2}. \tag{4}$$

The source-coefficient Stefan model uses $\hat{E}$ fitted on the other two regions, and the recalibrated Stefan model uses $\hat{E}$ fitted on the training labels of the target region. The physics-residual model retains the recalibrated Stefan prediction as an anchor and adds a learned correction (Fig. 2):

$$\widehat{\mathrm{ALT}} = \hat{E}\, s + \lambda\, g(\mathbf{x}), \qquad g = \arg\min_{f} \sum_{i=1}^{n} \left[\left(y_i - \hat{E}\, s_i \right) - f(\mathbf{x}_i) \right]^2, \tag{5}$$

where g is a CatBoost gradient-boosting regressor[31] (600 iterations, learning rate 0.03, depth 6, L2 regularisation 3) and $\lambda = 0.5$ shrinks the correction towards the physical anchor ($\lambda = 0.25$ and 1 were tested as sensitivity cases, and a value selected by inner block cross-validation with the full-season input). Three alternative structures place the physical information elsewhere,

$$\widehat{\mathrm{ALT}}_{\mathrm{D}} = f_{\mathrm{D}}(\mathbf{x}), \qquad \widehat{\mathrm{ALT}}_{\mathrm{I}} = f_{\mathrm{I}}\left(\mathbf{x},\, \hat{E}\, s,\, \widehat{\mathrm{ALT}}_{\mathrm{K}}\right), \qquad \widehat{\mathrm{ALT}}_{\mathrm{A}} = f_{\mathrm{A}}(\mathbf{x}) \text{ fitted to } \{(\mathbf{x}_i, y_i)\}_{i=1}^{n} \cup \{(\mathbf{x}_j, \hat{E}\, s_j)\}_{j=1}^{n}, \tag{6}$$

where direct ML (f_{D}) fits CatBoost to ALT from $\mathbf{x}$ alone, physics-input ML (f_{I}) adds the recalibrated Stefan prediction and the Kudryavtsev[32] prediction $\widehat{\mathrm{ALT}}_{\mathrm{K}}$ as features, and physics pseudo-labels (f_{A}) augment the n training rows with an equal number of rows j drawn from the cells of the scoring fold, with their covariates and the training-fold Stefan prediction as the target. All learned models were fitted with three random seeds and their predictions averaged (the transfer models without target labels with one seed). Random forest, a multilayer perceptron, TabM and FT-Transformer[33–35] replaced CatBoost in a learner sensitivity test, and SHAP values[36] of the residual and direct models were computed for models fitted to all labels of each region.

Validation designs and uncertainty

All models were evaluated on out-of-fold predictions under five designs[14,15]: random five-fold cross-validation by site-year; spatial-block five-fold cross-validation, with 0.5° blocks assigned to folds balanced by the number of site-years (main design); year hold-out, in which each observation year with at least 20 site-years is predicted by models trained on the other years; block-by-year hold-out, in which both the block fold and the year of the scored site-years are excluded from training (folds with at least 10

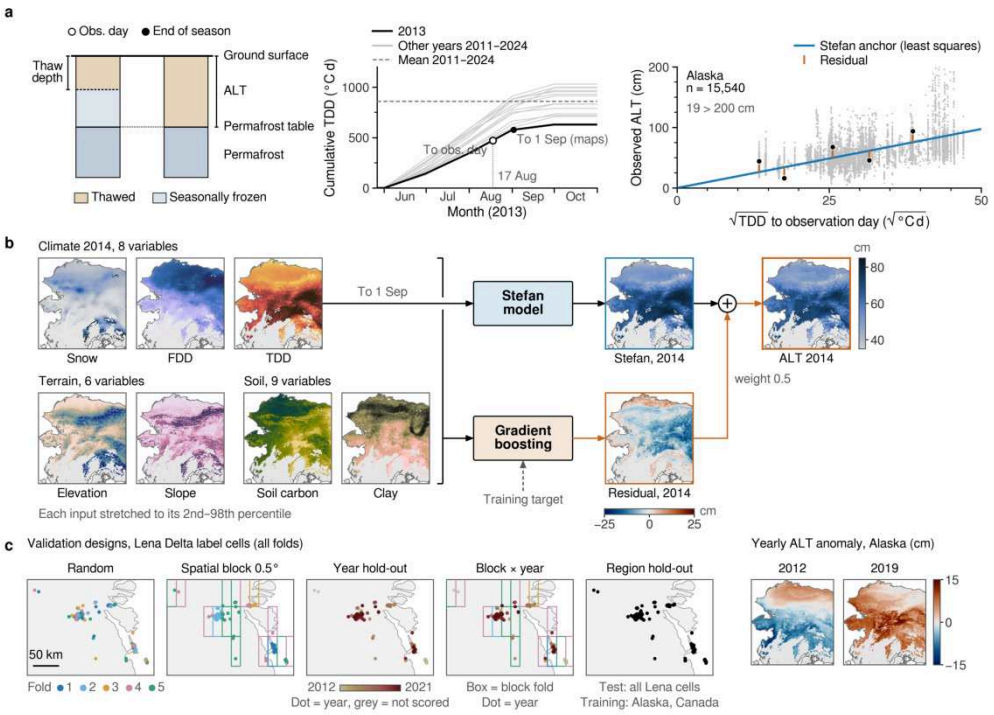


Fig. 2 Site-year design, Stefan anchor, physics-residual model and validation designs. **a**, Thaw depth on the observation day and ALT at the end of the season (left); cumulative thawing degree-days of a Lena Delta cell in 2013, with the model input accumulated to the observation day and the map input to 1 September (centre); observed ALT against $\sqrt{\mathrm{TDD}_{\mathrm{obs}}}$ for the Alaska site-years with the Stefan anchor and five residuals (right). **b**, Physics-residual model: the Stefan anchor plus gradient boosting of its residual from 23 climate, terrain and soil inputs (equation (5)), illustrated with the Alaska map of 2014. **c**, Validation designs on the Lena Delta label cells (colour marks the fold, the observation year or the test region), and ALT anomalies of the Alaska map in 2012 and 2019.

site-years); and region hold-out, in which a region is predicted by models trained on the other two regions after removing training cells within 100 km of the test region. Accuracy is expressed as

$$\mathrm{RMSE} = \left[\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2\right]^{1/2}, \qquad \Delta\mathrm{RMSE}_{A,B} = \mathrm{RMSE}_A - \mathrm{RMSE}_B, \tag{7}$$

where both terms of ΔRMSE are computed on the same site-years, so that negative values favour method A. Uncertainty was quantified with a paired block bootstrap: the 0.5° blocks were resampled with replacement $B = 2000$ times (1000 times for the label-number experiments and for part of the covariate-set and learner sensitivity runs, Supplementary Table S4), ΔRMSE was recomputed from all site-years of the resampled blocks, and the 2.5th and 97.5th percentiles of the replicates define the 95%

interval,

$$\mathrm{CI}_{95} = \left[Q_{0.025}\left(\Delta\mathrm{RMSE}^{(b)}\right),\ Q_{0.975}\left(\Delta\mathrm{RMSE}^{(b)}\right) \right], \quad b = 1, \ldots, B. \tag{8}$$

We describe a difference as resolved when its interval excludes zero. Conditions of the gain were examined within regional terciles of eight covariates (full-season TDD of the observation year, freezing degree-days, snow water equivalent, soil organic carbon, clay, slope, elevation and summer NDVI); terciles with fewer than 30 site-years or fewer than five blocks were not evaluated. Year-wise RMSE was computed for years with at least 30 site-years, and the correlation between observed and predicted year-to-year changes was computed at cells observed in several years.

Label-number and transfer experiments

Within each region, the training labels of each spatial-block fold were subsampled to n = 25, 50, 100, 250, 500, 1000, 2500 or all (three random draws), the Stefan coefficient was recalibrated on the same n labels, and the complete test fold was scored. For transfer, each region was scored in its spatial-block folds by models trained on all labels of the other two regions plus $n = 0$, 25, 50, 100, 250 or 500 labels drawn from the training folds of the target region; the Stefan coefficient was refitted on the n target labels, and the residual learner was trained on the source rows anchored with the source coefficient together with the target rows anchored with the refitted coefficient,

$$\widehat{\mathrm{ALT}}_{\mathrm{T}} = \hat{E}_{\mathrm{T}}\, s + \lambda\, g_{\mathrm{ST}}(\mathbf{x}), \qquad g_{\mathrm{ST}} = \arg\min_{f} \Big[\sum_{i \in \mathrm{S}} \left(y_i - \hat{E}_{\mathrm{S}}\, s_i - f(\mathbf{x}_i)\right)^2 + \sum_{i \in \mathrm{T}} \left(y_i - \hat{E}_{\mathrm{T}}\, s_i - f(\mathbf{x}_i)\right)^2 \Big], \tag{9}$$

where S denotes the source rows, T the n target rows, $\hat{E}_{\mathrm{S}}$ the coefficient fitted on the source rows and $\hat{E}_{\mathrm{T}}$ the coefficient refitted on the target rows ($\hat{E}_{\mathrm{T}} = \hat{E}_{\mathrm{S}}$ for $n < 3$); the source rows were not buffered by 100 km in this experiment. The external test used CALM observations of 2010–2024[5] outside the three study regions and outside mountain permafrost (Switzerland and Kazakhstan). Censored values, values flagged as early, burnt, flooded or not reaching the permafrost table, and sites within 100 km of the study labels were removed, leaving 816 site-years in 73 cells in eight regions (Supplementary Methods S3). Each external region was scored block by block (cell by cell for Greenland, which forms one block) by models trained on the three study regions plus $n = 0$, 5, 10, 25, 50 or all labels from the other blocks of the same region. One Svalbard borehole with a mean ALT of about 6 m, outside the training range, was excluded from scoring, which left 805 site-years in 72 cells and 49 blocks. The runs are listed in Supplementary Table S4.

Comparison with published products

Six products were sampled at the label cells: Wei et al.[13] (1 km, 2000–2024), Liu et al.[12] (1 km, 2003–2020), ESA CCI Permafrost ALT v4 (1997–2021) and v5 (1997–2023)[8,9], Yi and Kimball[29] (Alaska, 2001–2015) and Aalto et al.[10] (one value per cell for the 2000–2014 baseline). For yearly products the value of the observation year was

used (Supplementary Methods S4). Products and models were compared on common site-years, at which Wei et al., Liu et al. and ESA CCI v4 all provide a value for the observation year (2010–2020); ESA CCI v5 and Aalto et al. were scored on the common site-years where they have a value, and the year-wise comparison used all site-years of a year with a value of Wei et al. Wei et al. published the coordinates of their training sites, and a leakage mask removed label cells within 25 km of these sites. Liu et al. were trained on an earlier version of the ABoVE observations used here[12], so their Alaska and Canada values are not independent of our labels. To test whether the residual learner can also correct a product, the product value p of the observation year was recalibrated on the training fold,

$$\hat{p} = a + b\,p, \tag{10}$$

and the residual $y - \hat{p}$ was learned with the same covariates, learner, λ and folds as in equation (5) (Supplementary Methods S5).

Yearly mapping, area of applicability and prediction intervals

For each of the three regions, the recalibrated Stefan and physics-residual models were fitted to all site-years of the region and applied to the 1 km cells with an ESA CCI permafrost probability of at least 50%, excluding cells that are water (mean Global Surface Water occurrence of at least 50%) or lack an input (858,517, 38,086 and 1,151,169 cells), using the climate of each year from 2011 to 2024 with the TDD accumulated to 1 September. The area of applicability followed Meyer and Pebesma[37]. With $\mathbf{z}$ the standardised 23-dimensional feature vector (with the 2011–2024 mean of the yearly map inputs), the dissimilarity of map cell k is $d_k = \min_j \|\mathbf{z}_k - \mathbf{z}_j\|_2$ over all training rows j, and the cell is inside the area of applicability when

$$d_k \le \tau, \qquad \tau = Q_{0.95}\left\{ \min_{j:\, b(j)\neq b(i)} \|\mathbf{z}_i - \mathbf{z}_j\|_2 \;:\; i \in \text{training rows} \right\}, \tag{11}$$

where $b(\cdot)$ denotes the spatial block. For year y and cell k, the anomaly is $a_{k,y} = \widehat{\mathrm{ALT}}_{k,y} - \overline{\widehat{\mathrm{ALT}}_k}$, where the overbar denotes the 2011–2024 mean; the regional mean anomaly is the mean of $a_{k,y}$ over the mapped cells, and the trend of each cell is the least-squares slope

$$\beta_k = \frac{\sum_y (y - \bar{y})\, a_{k,y}}{\sum_y (y - \bar{y})^2}, \tag{12}$$

expressed in cm per decade. The observed anomaly of a year is the mean over the labels of that year of $y_i - \overline{\widehat{\mathrm{ALT}}_{k(i)}}$, where $k(i)$ is the map cell nearest to site i within 1.5 km, computed for years with at least 50 site-years and centred on the mean predicted anomaly of the same years. The 90% prediction interval of a map value is

$$\left[\widehat{\mathrm{ALT}}_{k,y} + Q_{0.05}(r),\ \widehat{\mathrm{ALT}}_{k,y} + Q_{0.95}(r) \right], \tag{13}$$

where r denotes the spatial-block out-of-fold residuals of the region. For display, the 1 km values were interpolated bilinearly onto a 250 m (Lena Delta) or 500 m (Alaska, Canada) hillshade grid. The map of 2013, the year with the most observed cells in Alaska and Canada among the years observed in all three regions, was compared with the two ML products on the cells observed in that year where all three maps have a value, using the product values of 2013 at the label sites, the cell means of the observations and our spatial-block out-of-fold predictions; the product maps are displayed on the cells of our map, and each region uses one colour range from the 5th–95th percentiles of the pooled values of the three maps inside the area of applicability.

Computation

Models were fitted with fixed random seeds on cloud nodes (Rescale; 96-core CPU nodes, and one NVIDIA A100 GPU node for the neural learners); data preparation, aggregation and figures were run on a local server (Supplementary Table S4).

Code availability

The code for data preparation, modelling, evaluation and figures is available at [repository and DOI]. Every number in the text is generated from the stored result tables by the scripts that produce the manuscript macros.

Results

Effect of the year-specific climate input

The observations are unevenly distributed in time (Fig. 1b, Table 1): 51% of the Alaska site-years were observed in 2014, whereas the Lena Delta labels are spread over 13 years. Most measurements were made in August (median day of year 224 in Alaska, 234 in the Lena Delta and 237 in Canada), before the end of the thaw season, so that the TDD accumulated to the observation day amounted to a median of 0.73, 0.81 and 0.89 of the full-season sum (equation (2)).

Replacing the mean climate input by the year-specific input reduced the spatial-block RMSE of the recalibrated Stefan model in the Lena Delta (ΔRMSE -1.4 [-2.6, -0.8] cm) and that of the physics-residual model in the Lena Delta (-1.6 [-2.4, -1.0] cm) and Canada (-1.5 [-2.0, -0.4] cm) (Fig. 3a). In Alaska the change was not resolved (Stefan $+1.8$ [-0.1, $+3.9$] cm), consistent with the concentration of the labels in one year, for which the year-specific input adds little beyond the spatial pattern. Beyond accuracy, the year-specific input ties each prediction to the climate of a stated year, which the yearly maps presented below require.

Accuracy under spatial, temporal and regional validation

Model rankings depended on the validation design (Fig. 3b, Supplementary Table S5). Random cross-validation gave the lowest errors to direct ML (11.8 cm in Alaska), and spatial-block cross-validation raised the direct-ML error by 5.4, 3.6 and 9.9 cm in

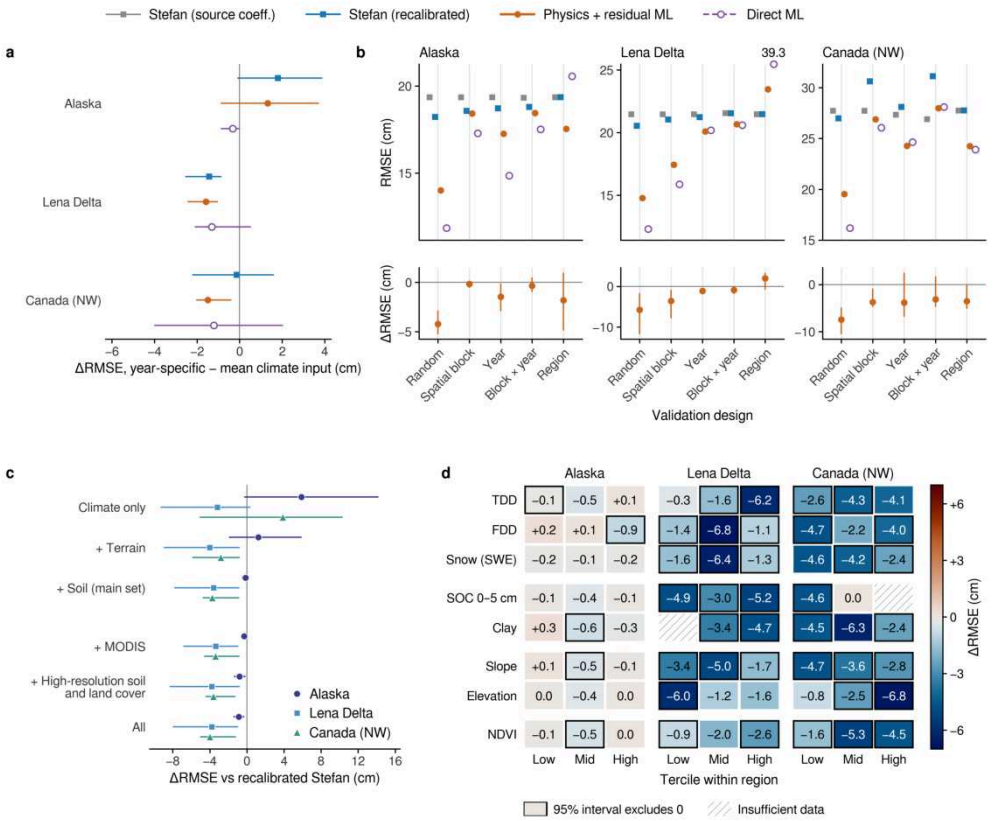


Fig. 3 Climate input, validation design and conditions of the physics-residual gain. a, Change in spatial-block RMSE when the 2011–2024 mean climate input is replaced by the year-specific input. **b**, RMSE of the four methods under five validation designs (upper; a value beyond the axis is printed) and the difference between the physics-residual model and the recalibrated Stefan model under each design (lower). **c**, The same difference for six covariate sets under spatial-block validation. **d**, The same difference within regional terciles of eight covariates; outlined cells have intervals excluding zero, and hatched cells have fewer than 30 site-years or five blocks. Intervals are 95% paired block-bootstrap intervals; negative values favour the first-named model.

Alaska, the Lena Delta and Canada. Year hold-out raised the direct-ML error in the Lena Delta from 15.9 to 20.2 cm, and the block-by-year design produced the largest within-region errors of all methods fitted to the labels of a region. When an entire region was withheld, the direct-ML error in the Lena Delta rose to 39.3 cm, against 21.5 cm for the Stefan model with the coefficient of the other regions (ΔRMSE +17.8 [+8.0, +23.9] cm); the physics-residual model stayed close to the Stefan model there (+2.0 [−0.8, +3.4] cm) and improved on it in Canada (−3.5 [−5.1, −0.1] cm). Spatial-block cross-validation is therefore the main design and region hold-out the test of transferability.

Gain of physics-residual learning and its controls

Under spatial-block cross-validation with the year-specific input, the physics-residual model reduced the RMSE of the recalibrated Stefan model from 21.0 to 17.4 cm in the Lena Delta (ΔRMSE −3.6 [−7.8, −0.8] cm) and from 30.6 to 26.9 cm in Canada (−3.7 [−4.8, −0.8] cm), whereas in Alaska the two models reached 18.6 and 18.4 cm (−0.2 [−0.6, +0.1] cm). The gain depended on the covariates available to the residual learner (Fig. 3c): with climate covariates only, the residual model did not improve on the Stefan model in Alaska (+5.9 [−0.3, +14.2] cm) or Canada (+3.9 [−5.1, +10.3] cm), and adding terrain and soil covariates produced the gains above. MODIS vegetation and snow timing or high-resolution soil and land-cover classes changed the result by less than 0.5 cm in the Lena Delta and Canada, and the latter added a small gain in Alaska (−0.8 [−1.4, −0.2] cm).

The result also depended on how the physical model entered the learner (Supplementary Fig. S2). Stefan pseudo-labels reproduced the Stefan model (Lena Delta −0.3 [−0.4, −0.1] cm), and the Stefan prediction as an input feature gave results similar to direct ML. With all labels of a region, direct ML reached lower point errors than the physics-residual model (Alaska 17.3 versus 18.4, Lena Delta 15.9 versus 17.4 and Canada 26.1 versus 26.9 cm), but its difference from the recalibrated Stefan model was resolved only in Canada (−4.6 [−6.2, −0.2] cm), and its error increased under region hold-out. Replacing CatBoost by random forest, a multilayer perceptron, TabM or FT-Transformer did not change the sign of the physics-residual gain in the Lena Delta and Canada.

Across regional terciles of eight covariates, the gain occurred in most conditions of the Lena Delta (19 of 23 terciles with intervals excluding zero) and Canada (17 of 23) and increased with the error of the Stefan model (correlation across terciles −0.99 in the Lena Delta and −0.79 in Canada; Fig. 3d). In the warmest TDD tercile of the Lena Delta the gain reached −6.2 [−12.0, −0.7] cm, whereas no tercile in Alaska changed by more than 0.9 cm. Terrain variables accounted for 50% of the mean absolute SHAP value of the residual model in the Lena Delta and soil variables for 39% in Canada (Supplementary Fig. S3).

Sample efficiency and transfer to new regions

With 25 training labels drawn from the site-years of a region, the physics-residual model was at least as accurate as a Stefan model recalibrated on the same labels (Alaska −0.9 [−1.9, +0.1], Lena Delta −0.8 [−2.2, −0.1] and Canada −0.3 [−3.2, +1.7] cm), and with 100 labels its gain was resolved in all three regions (−0.7 [−1.4, 0.0], −2.0 [−5.3, −0.3] and −1.9 [−3.7, −0.3] cm; Fig. 4a). Direct ML trained on 25 labels had a higher point error than the Stefan model in Alaska (+2.8 [−0.3, +4.5] cm) and Canada (+1.2 [−1.5, +3.4] cm). Physics-input ML and physics pseudo-labels followed the patterns of their full-data results (Supplementary Fig. S4).

In the transfer experiment, each region was predicted by models trained on the other two regions and n labels of the target region (Fig. 4b). Without target labels, direct ML reached 39.5 cm in the Lena Delta, against 21.5 cm for the source-coefficient Stefan model and 23.8 cm for the physics-residual model; 25 target labels reduced

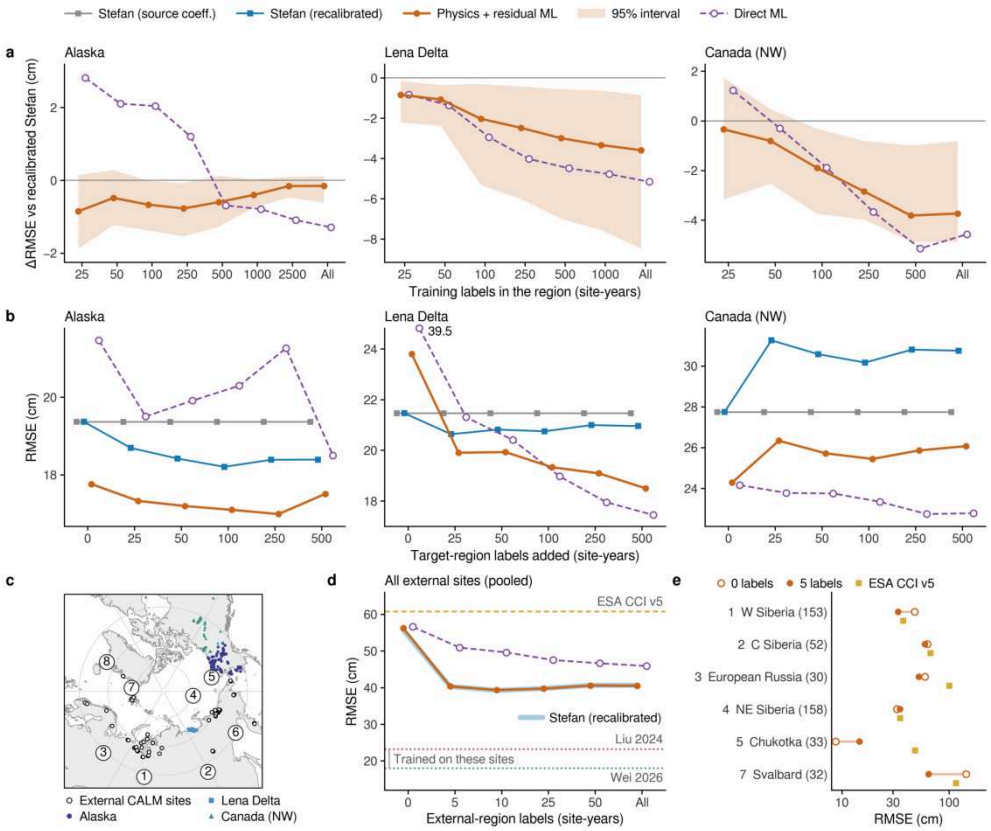


Fig. 4 Accuracy with few labels, in new regions and in external CALM regions. a, Difference in RMSE between each method and a Stefan model recalibrated on the same n training labels within each region (mean of three draws; band, 95% block-bootstrap interval of the physics-residual model). **b**, RMSE in each region for models trained on the other two regions plus n labels of the target region; a value beyond the axis is printed. **c**, External CALM sites numbered by sub-region, and the label cells of the study regions. **d**, RMSE in the external regions against the number of labels from the same region, on the 474 site-years with values of all three products; horizontal lines mark the products. **e**, RMSE of the physics-residual model with no and with five local labels, and of ESA CCI v5, by sub-region (site-years in brackets).

the physics-residual error to 19.9 cm and the direct-ML error to 21.3 cm. In Alaska the physics-residual model gave the lowest error at every n (17.8 cm without target labels). In Canada, direct ML trained on the source regions was the most accurate (24.2 cm); recalibrating the Stefan coefficient on 25 Canadian labels increased its error from 27.7 to 31.3 cm, and the physics-residual model, which uses this anchor, followed (24.3 to 26.3 cm), reflecting the wide range of thaw depths among the Canadian sites.

The same procedure was applied to eight external CALM regions with 805 site-years not used in model development (Fig. 4c–e, Supplementary Fig. S5). On the 474 site-years of seven regions with values of Wei et al., Liu et al. and ESA CCI v5, the physics-residual model trained on the three study regions reached 56.2 cm without

local labels; five labels from the same region reduced the error to 40.3 cm and 25 labels to 39.7 cm, whereas direct ML reached 47.5 cm with 25 labels. With five labels the physics-residual model was more accurate than ESA CCI v5, which is not trained on ALT observations (60.7 cm on the same site-years; ΔRMSE −19.5 [−31.3, −9.2] cm on all 700 site-years with a value). Wei et al. and Liu et al., which include these sites in their training data, reached 18.0 and 23.2 cm. Most of the reduction came from recalibrating the Stefan coefficient; the residual learner did not reduce the error further at these sites (+0.4 [−0.5, +1.2] cm with five labels), where the active layers of West and Central Siberia and Svalbard exceed the range of the study regions. The pooled curve combines different regional responses (Fig. 4e): with five labels the error decreased in West Siberia, Central Siberia, European Russia and Svalbard, but increased in Chukotka (8.8 to 14.6 cm), Kamchatka (7.9 to 11.7 cm; 16 site-years, not shown) and north-eastern Siberia (32.6 to 34.8 cm), where the error without local labels was already small.

Comparison with published products in the same cells and years

On the common site-years (2010–2020), the physics-residual model reached 21.1 cm in the Lena Delta, against 28.4 cm for ESA CCI v5, 30.6 cm for Aalto et al., 31.0 cm for ESA CCI v4, 33.0 cm for Liu et al. and 34.6 cm for Wei et al. (Fig. 5a, Supplementary Table S6). On the Lena Delta site-years more than 25 km from the training sites of Wei et al. (802 site-years with a value of Wei et al., 2,352 with a value of ESA CCI v5), the paired differences were −12.5 [−15.6, −5.8] cm relative to Wei et al. and −5.0 [−9.5, −2.5] cm relative to ESA CCI v5. In Alaska (17.1 versus 17.9 and 18.8 cm for Wei et al. and Liu et al.) and Canada (25.8 versus 27.0 and 27.9 cm), the model and the two ML products performed on par.

On the Lena Delta site-years, the two ML products assign most cells values between about 40 and 70 cm regardless of the observed ALT (Fig. 5b): the correlation between observed and predicted values was 0.05 for Wei et al. and 0.07 for Liu et al., against 0.73 for the physics-residual model, and the mean bias was +21.9 and +19.8 cm, against +2.8 cm. The bias of all series increased towards thin active layers (Fig. 5c): for site-years with an observed ALT below 30 cm (248 in the Lena Delta), Wei et al. and Liu et al. overestimated by +33.2 and +30.9 cm and the physics-residual model by +11.8 cm; in Alaska the same class (1,093 site-years) was overestimated by +19.5, +25.4 and +8.5 cm.

The regional contrast follows the training data of the products. Alaska label cells lie at a median distance of 7 km from the training sites of Wei et al., and Liu et al. were trained on an earlier version of the same Alaska and Canada field data, whereas the Lena Delta cells lie at a median distance of 107 km (Supplementary Fig. S6). Both products overestimate the thin active layers of the northern delta, with mean biases of +21.1 and +19.9 cm north of 72°N. A model trained without any Lena Delta label was already more accurate there than Wei et al. on the same site-years (26.0 versus 30.3 cm; ΔRMSE −4.3 [−6.1, −0.8] cm), so that the advantage in this region does not depend on local labels alone.

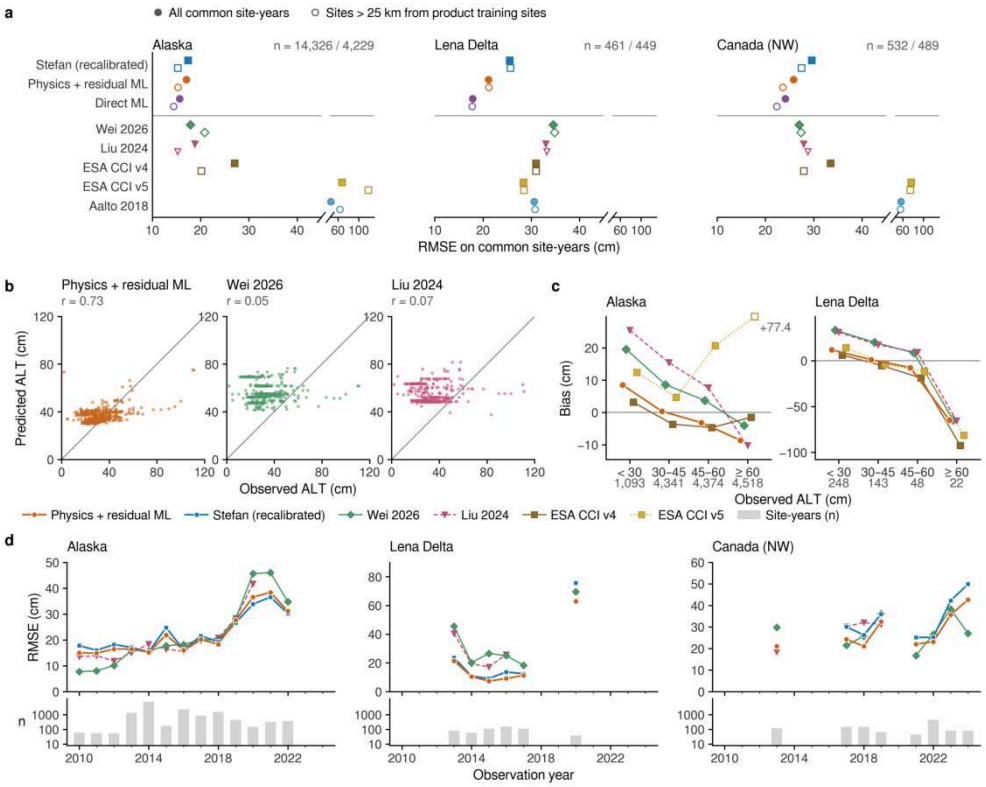


Fig. 5 Comparison with published products on the same cells and years. a, RMSE of three methods and five products on the common site-years (2010–2020), for all cells (filled) and for cells more than 25 km from the training sites of Wei et al. (open); values above 45 cm are drawn on the compressed segment. **b**, Observed against predicted ALT on the 461 common site-years of the Lena Delta, with the 1:1 line and the Pearson correlation. **c**, Mean bias within classes of observed ALT (site-years per class below the axis; classes with fewer than 10 site-years omitted). **d**, RMSE per observation year for years with at least 30 site-years, with the number of site-years per year. Our methods use spatial-block out-of-fold predictions throughout.

Maps of 2013, the year with the most observed cells in Alaska and Canada among the years observed in all three regions, show these differences in space (Fig. 6). At the 126 Alaska cells observed in that year, the RMSE of the three maps was similar (14.9, 13.4 and 12.9 cm), whereas the correlation with the observations was lower for our model (0.34) than for the products (0.53 and 0.57), 59 of these cells lying within 25 km of a training site of Wei et al. The largest differences between the maps occur above 600 m in mountain terrain; between 68 and 70°N the products are deeper than our map there by +18.7 and +52.9 cm. In the Lena Delta both products were deeper than the observations (mean bias +19.8 and +20.7 cm; RMSE 32.3 and 29.6 cm against 16.6 cm), although only 6 cells were observed in 2013. In Canada (NW), the RMSE at the 33 observed cells was 23.7 cm for our model, 29.9 cm for Wei et al. (mean bias +20.5 cm) and 20.2 cm for Liu et al., whose training data include an earlier version of these observations.

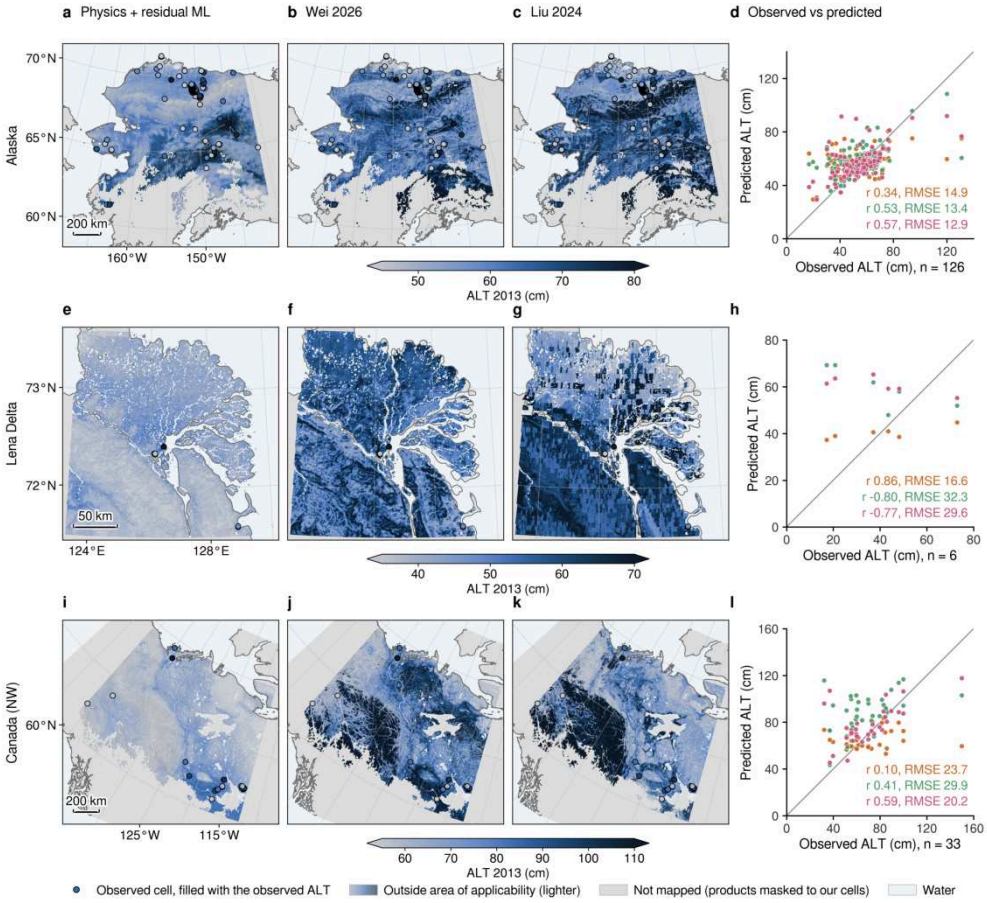


Fig. 6 ALT maps of 2013 from the physics-residual model and two published products, with the observed values. a–c, Alaska: physics-residual model, Wei et al. and Liu et al.; the product maps are shown on the cells of our map, and the cells observed in 2013 are drawn as circles filled with the observed ALT on the same colour scale. **d**, Observed against predicted ALT at the 126 observed cells with values in all three maps, with the 1:1 line, the correlation and the RMSE; our predictions are spatial-block out-of-fold values. **e–h**, The same for the Lena Delta (6 cells). **i–l**, The same for Canada (NW) (33 cells). Each region has one colour range from the 5th–95th percentiles of the three maps inside the area of applicability; lighter colours mark cells outside it.

Year by year, the physics-residual model had a lower error than both Wei et al. and the Stefan model in all 6 Lena Delta years with at least 30 site-years (2013–2017 and 2020), in 7 of 13 Alaska years against Wei et al. and 9 of 13 against the Stefan model (2010–2022), and in 7 of 8 Canadian years against the Stefan model (Fig. 5d). At Alaska cells observed in several years, the correlation between observed and predicted year-to-year changes was 0.28 for the physics-residual model and 0.15 and 0.12 for Wei et al. and Liu et al., whereas in the Lena Delta the products followed the interannual changes more closely (0.55 and 0.42 versus 0.29).

The residual learner can also be applied with a product as the anchor (equation (10)). With the Lena Delta labels, adding the residual learner to the recalibrated product of Wei et al. changed its error by −2.3 [−7.4, −0.4] cm, and the Stefan-anchored model remained more accurate than the product-anchored model (−2.8 [−6.2, −0.1] cm; Supplementary Fig. S7); in Alaska and Canada the two anchors were not distinguishable.

Year-labelled 1 km ALT maps, 2011–2024

The physics-residual models fitted to all site-years of each region were applied with the ERA5-Land climate of each year from 2011 to 2024 (Fig. 7); the maps cover 858,517 cells in Alaska, 38,086 in the Lena Delta and 1,151,169 in Canada (NW). Relative to the 2011–2024 mean, the regional mean anomaly ranged from −2.8 cm in 2011 to +6.0 cm in 2019 in Alaska, from −8.9 cm in 2016 to +9.0 cm in 2020 in the Lena Delta and from −3.8 cm in 2020 to +6.4 cm in 2023 in Canada, with spatial patterns that differ between years (Fig. 7f). In the Lena Delta, the mean anomaly of the observed cells had the same sign as the predicted regional anomaly in 8 of 10 years with at least 50 site-years (2012–2021; $r = 0.73$), and the correlation remained 0.75 without 2020, when the observed anomaly of 143 site-years reached +46.5 cm. In Alaska, where the set of observed cells changes between campaigns, the agreement was weaker ($r = 0.13$) and the predicted interannual range (8.8 cm) was smaller than that of the observed cell means (17.8 cm); in Canada the correlation over 7 years was 0.64, with the observed anomalies of 2023 and 2024 (80 site-years each) far above the predicted range (Fig. 7d). The linear trend of the yearly maps averaged +2.1, +6.7 and +1.5 cm per decade over the Alaska, Lena Delta and Canada cells, with positive values in 78%, 100% and 91% of the cells (Fig. 7e); it reflects the ERA5-Land climate of the period as transmitted by the model and is not an observed trend. The area of applicability covers 56% of the Alaska cells, 62% of the Lena Delta cells and 43% of the Canada cells; cells outside it lie mainly in mountains and at the southern margin of the permafrost zone (Supplementary Fig. S8). The half-width of the 90% prediction interval is 25.5, 24.5 and 41.3 cm. The correction added by the residual learner is mostly within ±10 cm (Supplementary Fig. S9), and series at long-record sites are shown in Supplementary Fig. S10.

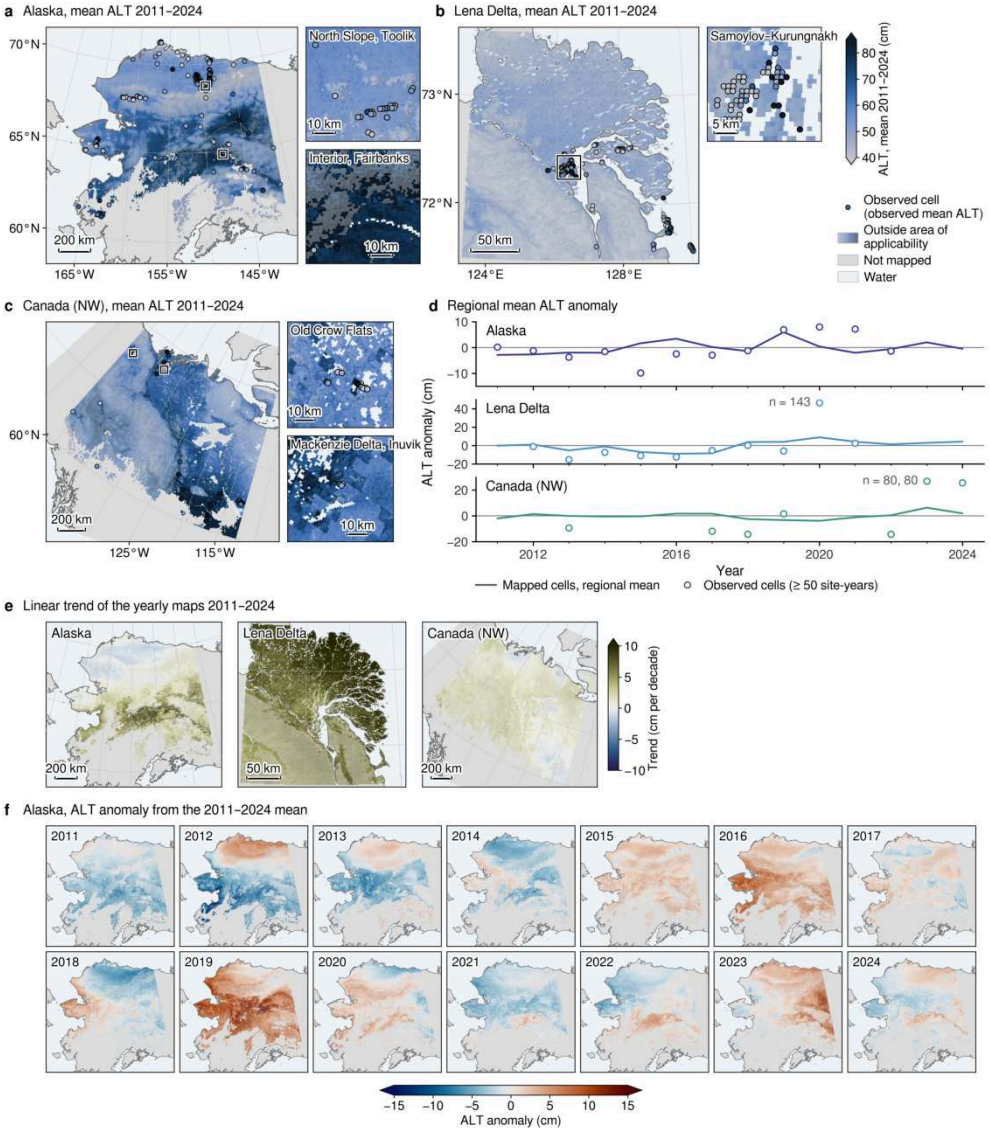


Fig. 7 Year-labelled 1 km ALT maps, 2011–2024. a–c, Mean ALT of 2011–2024 from the yearly maps of Alaska, the Lena Delta and Canada (NW), with windows enlarged at the right (1 km cells) and the observed cells filled with their observed mean ALT; lighter colours mark cells outside the area of applicability. **d**, Regional mean anomaly of the maps and mean anomaly of the observed cells in years with at least 50 site-years, centred on the predicted anomaly of the same years (y ranges differ between regions). **e**, Linear trend of the yearly maps (least squares per cell), which describes the modelled response to the climate of each year. **f**, ALT anomaly of each year relative to the 2011–2024 mean in Alaska.

Discussion

The way physical knowledge enters the model determined whether machine learning improved on a regional Stefan model and whether the improvement survived the transfer to a region without labels. In the physics-residual formulation (equation (5)), the Stefan term carries the response to the thaw-season climate of each year in a bounded form, and the learner corrects the departures of the effective coefficient that soil and terrain impose. Four results support this division of roles: the gain required terrain and soil covariates and did not appear with climate covariates alone; it was largest where the Stefan error was largest; the attribution of the residual learner pointed to terrain in the Lena Delta and to soil organic carbon in Canada; and Stefan pseudo-labels pulled the learner towards the Stefan model without adding accuracy. Direct ML, which has to learn the climate response from the labels, reached lower point errors when all labels of a region were available, but its Lena Delta error exceeded that of the Stefan model by 17.8 cm when the region was withheld, and with 25 labels it was less accurate than the Stefan model in Alaska and Canada. The gain of the residual learner is therefore conditional, and the conditions are themselves a result: it is worth adding a learner to the physics where soil and terrain information is available and where the regional Stefan coefficient misrepresents the sites, and it is worth anchoring the learner to the physics where the target region has few labels, which is the common situation in permafrost regions.

The comparison with the published products on the same cells and years explains why their accuracy differs between regions. The two ML products relate site observations pooled over many years to multi-year predictors, and their validation uses random or site-based splits on their own site sets[12,13]. Where their training sites are close to our label cells (Alaska, and Canada for Liu et al., whose training data include an earlier version of the same field observations), the products were as accurate as our model, and the comparison there is not independent. Where the products have no nearby training sites (the Lena Delta, median distance 107 km), they assigned values between about 40 and 70 cm regardless of the observed ALT (correlation 0.05 and 0.07 against 0.73), overestimated the thin active layers of the northern delta by about 20 cm, and were less accurate than a physics-residual model trained without any Lena Delta label. Three properties of our workflow produce this difference: the climate input of each prediction is the thawing degree-day sum of the stated year accumulated to the observation day, so that a thin active layer in a cold year is not averaged with thick ones; the Stefan anchor transfers the climate response to cells and years outside the training set; and the evaluation scores every method and product on the same site-years with a mask for product training sites, which is what makes the regional ranking interpretable. The process-based ESA CCI product, which is not trained on ALT observations, had larger errors than our model with five local labels at the external CALM sites, which places the two approaches on one scale for the first time.

These results extend previous comparisons in three directions. Gautam et al.[23] found at 68 Alaska CALM sites of one year that a random forest lost most of its accuracy on held-out sites whereas the Stefan model retained it; we confirm this ordering over site-years of 2010–2024 in three regions and under region hold-out, and

we show that the two approaches need not be alternatives, because a learner that corrects the Stefan model retains its transferability. Studies of physics-guided learning in lakes and rivers reported that process knowledge mainly helps under data scarcity and extrapolation[20–22]; the label-number and region hold-out experiments give this statement a quantitative form for ALT, with the number of labels at which the gain becomes resolved and the error without target-region labels. Finally, the finding that random cross-validation underestimates the spatial-block error of direct ML by 3.6–9.9 cm, and that the region hold-out error is larger still, follows the literature on spatial validation[14–16]; its consequence for ALT products is that errors from random or site-based splits do not indicate the accuracy of a product in a region without training observations, and that such regions are where products are most needed.

The yearly maps turn these results into a product with stated conditions of use. Each map is driven by the ERA5-Land climate of one year with the TDD accumulated to 1 September, so that it can be compared with observations or products of the same year, and it carries an area of applicability and a prediction interval. The regional mean anomalies of the maps followed the observed anomalies in the Lena Delta (r = 0.73), where the observation network is stable, but only weakly in Alaska (r = 0.13), where the set of observed cells changes between campaigns and the observed range of regional anomalies therefore exceeds the predicted one; the linear trends of the maps describe the response of the model to the climate of 2011–2024 and are not observed trends. For a new region, the label-number experiments indicate what to expect rather than a fixed procedure. Without local labels, the Stefan model with the coefficient of other regions, or the physics-residual model trained on other regions, is the safer starting point. Local labels improve the model only when they represent the local range of soils and terrain: 25 Lena Delta labels reduced the error of the physics-residual model from 23.8 to 19.9 cm, and five labels reduced the external CALM error from 56.2 to 40.3 cm, whereas 25 Canadian labels increased the error of the recalibrated Stefan model from 27.7 to 31.3 cm because the thaw depths of the Canadian sites span a wide range. The spatial-block evaluation within the new region and the area of applicability of the resulting map are the two checks that distinguish these cases, and both are part of the workflow.

The study has limitations that bound these conclusions. Half of the Alaska labels come from 2014, which prevented a resolved estimate of the effect of the year-specific input in that region, and the Lena Delta labels come from one data set in 16 blocks, which widens the confidence intervals. The labels are observation sites rather than independent 1 km cells; several sites of one transect can lie in one cell, and although the spatial-block design removes the resulting dependence from the validation, the site-year counts overstate the number of independent samples at the map scale. The climate input is monthly and the soil covariates are static, so that short-term weather and changes in soil moisture are not represented. Active layers thicker than about 3 m, mainly in mountains and at the southern margin of the permafrost zone, lie outside the training range, and the external tests showed that the residual learner does not transfer to such conditions without local labels. The comparison of maps with observations in one common year rests on few cells in the Lena Delta and Canada.

Daily climate inputs, remotely sensed snow and soil-moisture information, and labels from a fourth region such as West Siberia, where CALM and campaign data both exist, are the next steps. Because every number in this study is tied to a stated site, year and validation design, the same protocol can be applied to new products and regions as they appear, and the year-labelled maps can be extended year by year as the climate data are released.

Conclusions

This study evaluated and mapped active-layer thickness in a site-year design, in which every label, climate input, prediction and score refers to one observation site in one stated year. Three findings are specific to this study. First, a Stefan model recalibrated with local labels and corrected by a gradient-boosting model of its residual reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada under spatial-block validation, and the gain was conditional: it required soil and terrain covariates and was largest where the regional Stefan coefficient misrepresented the sites. Second, the physical anchor preserved transferability: without labels of the target region the physics-residual model stayed close to the Stefan model, whereas direct ML exceeded the Stefan error by 17.8 cm in the Lena Delta, and 5–25 local labels were enough to reduce the physics-residual error in the Lena Delta and in external CALM regions. Third, scoring all models and five published products on the same cells and years showed that the accuracy of the ML products follows the location of their training sites; in the Lena Delta they did not represent thin active layers (correlation 0.05 and 0.07 with the observations) and were less accurate than a model trained without any label of the region. The workflow delivers year-labelled 1 km maps for 2011–2024 for three regions, each with an area of applicability and a prediction interval, together with the evaluation protocol and the label-number evidence needed to judge whether, and with how many labels, the method can be applied to another permafrost region.

Data availability. All input data are publicly available. ALT observations: ABoVE soil moisture and active-layer thickness, version 2 (ORNL DAAC, https://doi.org/10.3334/ORNLDAAC/2369), ALLena thaw-depth data (PANGAEA, https://doi.org/10.1594/PANGAEA.973813) and GTN-P CALM (PANGAEA, https://doi.org/10.1594/PANGAEA.972777). Climate: ERA5-Land monthly averaged data (Copernicus Climate Data Store). Covariates: Copernicus DEM GLO-30, SoilGrids 2.0, ESA CCI permafrost probability, and MODIS MOD13Q1 and MOD10A2 version 6.1 (accessed through AppEEARS). Products: Wei et al. (https://doi.org/10.5281/zenodo.21667583), Liu et al. (https://doi.org/10.5281/zenodo.10070610), ESA CCI ALT v4 and v5 (CEDA), Aalto et al. (https://zenodo.org/records/5003804) and Yi and Kimball (https://doi.org/10.3334/ORNLDAAC/1760). The site-year label table, the yearly climate-input table, the static covariate table, the out-of-fold predictions of all methods and the yearly 1 km maps of Alaska, the Lena Delta and northwestern Canada (2011–2024) will be deposited in a public repository upon acceptance [repository and DOI].

References

[1] Hjort, J. *et al.* Degrading permafrost puts Arctic infrastructure at risk by mid-century. *Nature Communications* **9**, 5147 (2018).

[2] Schuur, E. A. G., McGuire, A. D., Schädel, C. *et al.* Climate change and the permafrost carbon feedback. *Nature* **520**, 171–179 (2015).

[3] Biskaborn, B. K. *et al.* Permafrost is warming at a global scale. *Nat. Commun.* **10**, 264 (2019).

[4] Nelson, F. E. *et al.* Estimating active-layer thickness over a large region: Kuparuk River basin, Alaska, U.S.A. *Arct. Alp. Res.* **29**, 367–378 (1997).

[5] Streletskiy, D. A. *et al.* GTN-P CALM: 35 years of Active Layer Thickness (ALT) across latitudinal and elevational gradients in the Northern Hemisphere. PANGAEA (2025). URL https://doi.org/10.1594/PANGAEA.972777.

[6] Moore, M. A. *et al.* ABoVE: Soil moisture and active layer thickness in Alaska, USA and Canada, 2005–2024, version 2. ORNL Distributed Active Archive Center (2025). URL https://doi.org/10.3334/ORNLDAAC/2369.

[7] Veremeeva, A. *et al.* ALLena: Thaw depth measurements of the active layer in the Lena River Delta region from 1998 to 2022, Northeastern Siberia. PANGAEA (2025). URL https://doi.org/10.1594/PANGAEA.973813.

[8] Westermann, S. *et al.* ESA Permafrost Climate Change Initiative (Permafrost_cci): Permafrost active layer thickness for the Northern Hemisphere, v4.0. NERC EDS Centre for Environmental Data Analysis (2024). URL https://doi.org/10.5285/d34330ce3f604e368c06d76de1987ce5.

[9] Westermann, S. *et al.* ESA Permafrost Climate Change Initiative (Permafrost_cci): Permafrost active layer thickness for the Northern Hemisphere, v5.0. NERC EDS Centre for Environmental Data Analysis (2025).

[10] Aalto, J., Karjalainen, O., Hjort, J. & Luoto, M. Statistical forecasting of current and future circum-Arctic ground temperatures and active layer thickness. *Geophys. Res. Lett.* **45**, 4889–4898 (2018).

[11] Ran, Y. *et al.* New high-resolution estimates of the permafrost thermal state and hydrothermal conditions over the Northern Hemisphere. *Earth Syst. Sci. Data* **14**, 865–884 (2022).

[12] Liu, Y. *et al.* A 1 km annual active layer thickness dataset for the Northern Hemisphere permafrost region, 2003–2020. *Environmental Research Letters* **19** (2024). Fields to verify before submission.

[13] Wei, Y., Wu, Z., Wang, J. *et al.* A 1 km resolution dataset of Northern Hemisphere permafrost active layer thickness (2000–2024). *Earth System Science Data Discussions* (2026).

[14] Roberts, D. R. *et al.* Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **40**, 913–929 (2017).

[15] Ploton, P. *et al.* Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nat. Commun.* **11**, 4540 (2020).

[16] Meyer, H. & Pebesma, E. Machine learning-based global maps of ecological variables and the challenge of assessing them. *Nature Communications* **13**, 2208 (2022).

[17] Wadoux, A. M. J.-C., Heuvelink, G. B. M., de Bruin, S. & Brus, D. J. Spatial cross-validation is not the right way to evaluate map accuracy. *Ecol. Model.* **457**, 109692 (2021).

[18] Stefan, J. Über die Theorie der Eisbildung, insbesondere über die Eisbildung im Polarmeere. *Annalen der Physik* **278**, 269–286 (1891).

[19] Riseborough, D., Shiklomanov, N., Etzelmüller, B., Gruber, S. & Marchenko, S. Recent advances in permafrost modelling. *Permafrost and Periglacial Processes* **19**, 137–156 (2008).

[20] Willard, J., Jia, X., Xu, S., Steinbach, M. & Kumar, V. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Comput. Surv.* **55**, 66 (2022).

[21] Read, J. S. *et al.* Process-guided deep learning predictions of lake water temperature. *Water Resour. Res.* **55**, 9173–9190 (2019).

[22] Jia, X. *et al.* Physics-guided machine learning for scientific discovery: an application in simulating lake temperature profiles. *ACM/IMS Trans. Data Sci.* **2**, 20 (2021).

[23] Gautam, S., Mishra, U., Scott, S. N. & Lara, M. J. Machine learning and process-based modeling of spatiotemporal changes in active layer thickness across Alaska. *Sci. Rep.* **15**, 42420 (2025).

[24] Muñoz-Sabater, J. *et al.* ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* **13**, 4349–4383 (2021).

[25] European Space Agency. Copernicus global digital elevation model (GLO-30). Distributed by OpenTopography (2021).

[26] Poggio, L. *et al.* SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL* **7**, 217–240 (2021).

[27] Didan, K. MODIS/Terra vegetation indices 16-day L3 global 250m SIN grid V061. NASA EOSDIS Land Processes DAAC (2021).

[28] Hall, D. K. & Riggs, G. A. MODIS/Terra snow cover 8-day L3 global 500m SIN grid, version 61. NASA National Snow and Ice Data Center DAAC (2021).

[29] Yi, Y. & Kimball, J. S. ABoVE: Active layer thickness from remote sensing permafrost model, Alaska, 2001–2015. ORNL DAAC, Oak Ridge, Tennessee, USA (2019).

[30] Palmtag, J. *et al.* Detailed pedon data on soil carbon and nitrogen for the northern permafrost region, version 1. Bolin Centre Database (2022). URL https://doi.org/10.17043/palmtag-2022-pedon-1.

[31] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V. & Gulin, A. CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems* **31**, 6639–6649 (2018).

[32] Kudryavtsev, V. A., Garagula, L. S., Kondrat'yeva, K. A. & Melamed, V. G. *Osnovy Merzlotnogo Prognoza [Fundamentals of Frost Forecasting in Geological Engineering Investigations]* (Moscow State University, 1974). In Russian. English translation: CRREL Draft Translation 606, US Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1977.

[33] Gorishniy, Y., Rubachev, I., Khrulkov, V. & Babenko, A. Revisiting deep learning models for tabular data. *Advances in Neural Information Processing Systems* **34**, 18932–18943 (2021).

[34] Gorishniy, Y., Kotelnikov, A. & Babenko, A. TabM: advancing tabular deep learning with parameter-efficient ensembling. *International Conference on Learning Representations* (2025).

[35] Holzmüller, D., Grinsztajn, L. & Steinwart, I. Better by default: strong pre-tuned MLPs and boosted trees on tabular data. *Advances in Neural Information Processing Systems* **37** (2024).

[36] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*, Vol. 30, 4765–4774 (2017).

[37] Meyer, H. & Pebesma, E. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* **12**, 1620–1633 (2021).

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